AI CLUSTERS

Evolution of Congestion Control in AI Ethernet Fabrics

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Agenda

- Characteristics of AI Workload
- What drives traffic congestion
- Requirements of next-gen congestion control
- Emerging Congestion Control Algorithms
  - Receiver Credit-Based Congestion Control (RCCC)
  - Network Signaled Congestion Control (NSCC)
- Network Telemetry for Congestion Management

Characteristics of AI Workload

- Synchronized traffic patterns
- Burstiness
- Severe Incast
- Elephant flows
- Low entropy
- Requires High bandwidth and low latency
- Sensitivity to Tail latency

What drives traffic congestion?

- **Traffic pattern**: incast, all-to-all, many-to-one, synchronized collectives
- **Load imbalance**: hotspots, uneven flow distribution, elephant vs mice flows
- **Bandwidth mismatch**: NIC rate vs switch links, uplink/downlink ratio
- **Buffering**: Queue buildup, how deep are the queues
- **Latency and synchronization**: small delays can amplify into large bursts
- **Failures and rerouting**: reduced capacity can lead to transient congestion
- **Packets/flow behavior**: flow size, flow arrival timing

Where should congestion be detected, can the system react quickly, and where should the traffic be buffered?

Congestion points

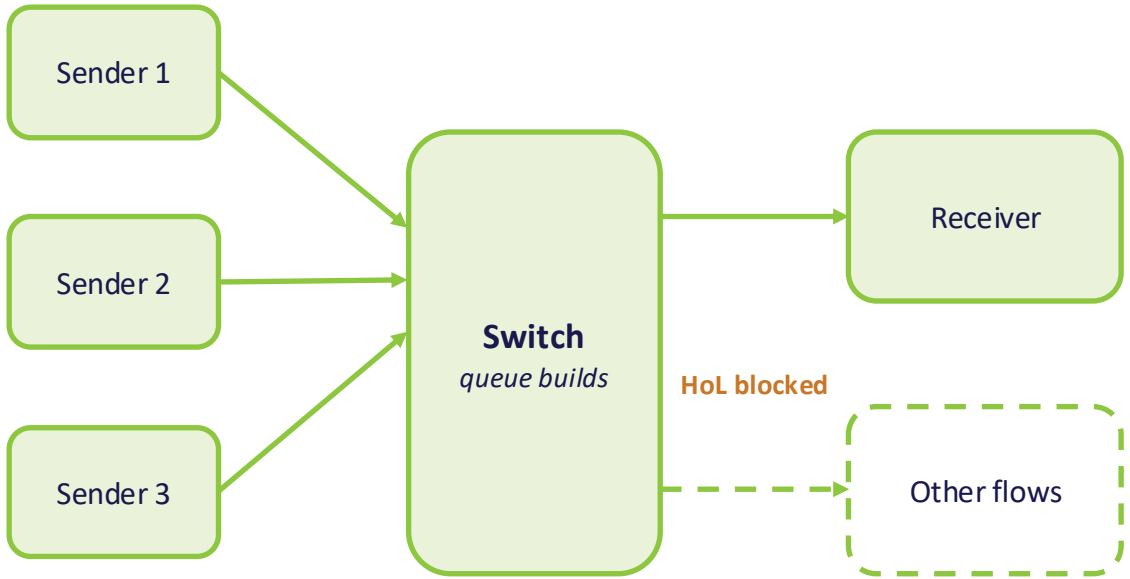
- NIC
- Switch
- PCIE Link



In context of network congestion, let's focus on the first two.

DCQCN - Congestion management for traditional AI networks

- Relies on ECN and PFC signals
- PFC causes congestion spread in the network and Head of Line Blocking
- Hard to Tune
- Significant Performance limitations
- Not Standardized – Not interoperable



PFC PAUSE stops the whole link, spreading congestion upstream

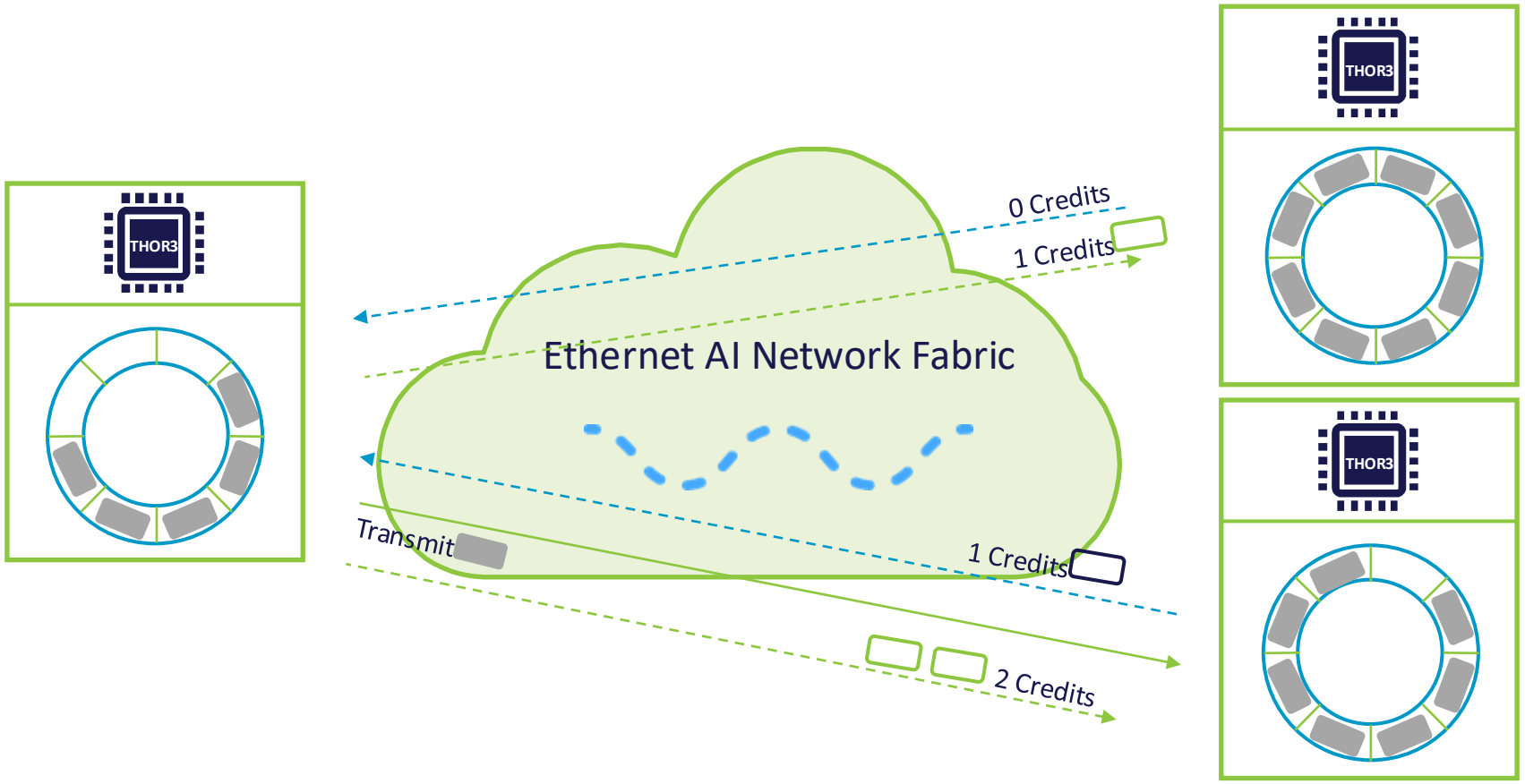
Requirements of congestion management for modern AI data center?

- Support multi-path
- Resilient to failure
- Scale without performance-tradeoff
- NIC flow control and network congestion signals should work in harmony
- Handle incast

Receiver Credit-Based Congestion Control (RCCC)

- Lossy network => no PFC
- Optimized for fully-provisioned networks running AI workloads
- Flow completion time close to optimal for AI traffic patterns
- RCCC operation highlights:
  - Line rate flow startup –optimal behavior for short flows
  - Receiver issues credit to active senders after 1st RTT
  - One-RTT convergence
  - ECN is used as an indicator of network congestion buildup
  - In steady state, last hop queues = 0 and full utilization.
  - Trimming is used to cope network buffer drops:
    - Receiver sends NACK for each trimmed packet.
    - Sender retransmits trimmed packets with priority when it has credits.

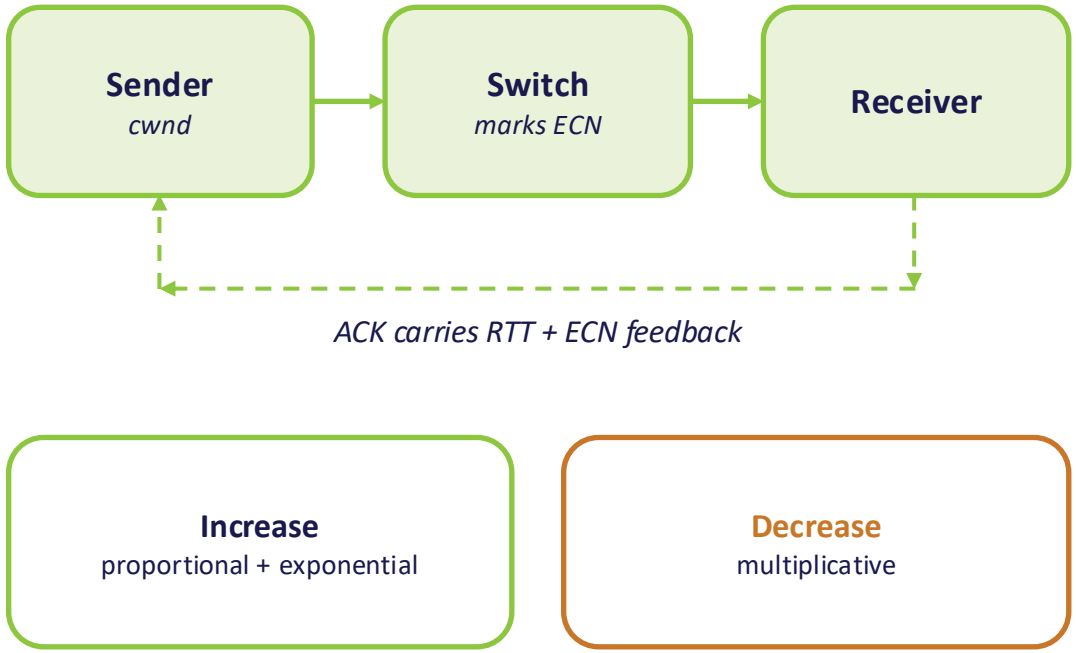
Receiver Based Congestion Control (RCCC)



Receiver NIC provides credits to Transmitter for Simple & Scalable CC

Network Signaled Congestion Control (NSCC)

- Sender-based algorithm using RTT and ECN to control window size
  - ECN is used as a leading indicator of switch queue congestion
  - Uses delay as primary measure of congestion
- A congestion window limits the amount of outstanding inflight data
- A combination of proportional and exponential algorithms are used to increase the congestion window
- A Multiplicative decrease is used to decrease the congestion window



Network Signals for Congestion Management

- Explicit Congestion Notification (ECN)
- RTT
- Packet Trimming
- CSIG

ECN
Switch marks packets as queues build

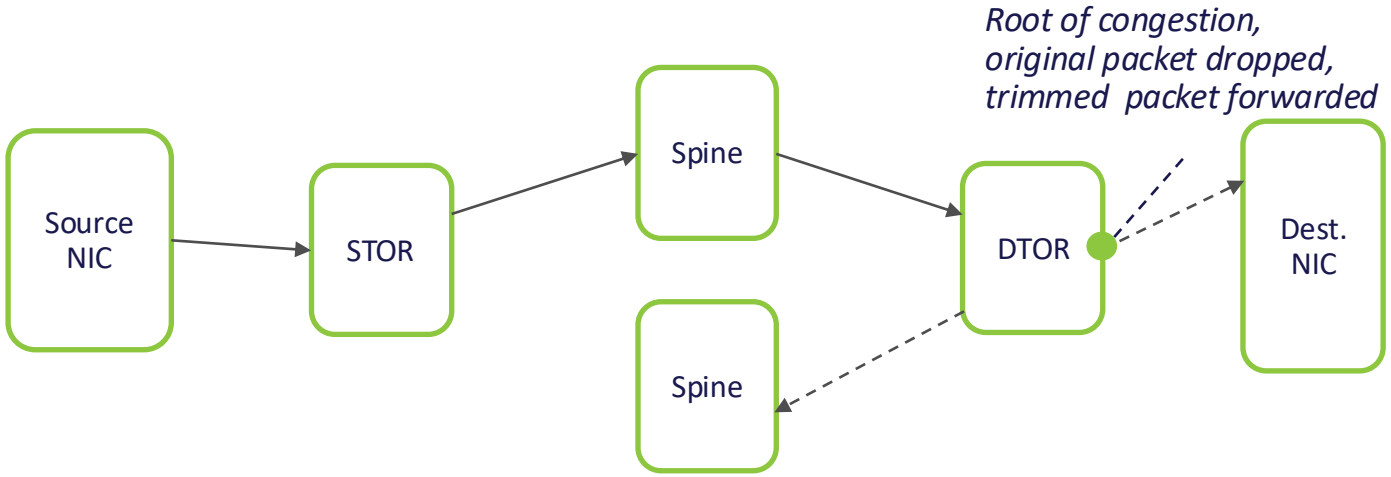
RTT
End-to-end delay measured by the sender

Packet Trimming
Header-only packet signals congestion instead of a drop

CSIG
In-band, per-packet telemetry from every hop

Packet Trimming

- Upon congestion, instead of dropping, transit switch sends truncated packet to the destination
- Trimmed packet is forwarded in a higher priority queue and has its DSCP rewritten
- Upon receiving a trimmed packet, the receiver generates a NACK



CSIG (Congestion Signaling)

- CSig is a new mechanism to gather packet level telemetry in-band without impacting the packet forwarding or the NOS
- The following telemetry (aka signals) gathered via per-packet fixed metadata
  - Minimum available bandwidth: Minimum available BW across all the ports in a packet path
  - Minimum available capacity: Minimum available capacity (%) across all the ports in a packet path
  - Max hop delay: Maximum delay observed across all the hops in a packet path
- Enables scalable congestion monitoring through hardware offloaded mechanism
- CSIG is being standardized in the UEC's Transport Working Group (TR WG)

Call to Action

- Learn about MRC at OCP Academy
- Help us understand what are unique challenges you face with your AI workloads.
- Use different congestion control with your collectives. Discuss which one works better?


Thank
You!
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